Part E — Written Justification

Week 3 Data Visualisation Assignment Roll No: BSAI-010

Panel 1 — Systolic BP Histogram

Question answered: What is the distribution of systolic blood pressure across all 10 000 admissions?
Chart type: Histogram
Week 2 channel rule followed: Position on a common scale (Lecture 4, Cleveland & McGill hierarchy, rank 1) is the highest-accuracy channel for communicating quantity. A histogram uses position (bar height) on a common vertical axis, making it optimal for showing distributional shape, skewness, and outliers without imposing a parametric assumption.
Parameters chosen and defended:

Parameter	Value	Reason
`bin_width`	5 mmHg	Clinical intervals in BP measurement are typically 5–10 mmHg; too narrow (e.g., 1 mmHg) amplifies noise, too wide (e.g., 20 mmHg) hides the skew.
`edgecolor`	white	Separates bars visually without adding heavy ink; avoids bars merging at low counts.
`color`	Okabe-Ito blue `#0072B2`	Categorical colour is unnecessary for a single series; blue is the primary course slot.

Exclusion: 0 rows removed for systolic (all values are in physiological range). The heart_rate > 300 removal (see below) did not affect systolic distribution materially (before mean: compare separately).

Panel 2 — Length of Stay by Ward

Question answered: Does length of stay differ across ward types?
Chart type: Box plot
Week 2 channel rule followed: Position on a common scale for the continuous variable (LOS), separated by a categorical variable (ward). A box plot efficiently summarises the 5-number summary (min, Q1, median, Q3, max) and makes ward-to-ward comparisons easy. Colour encodes ward identity as a redundant second channel (each ward gets a distinct hue from the Okabe-Ito set).
Parameters chosen and defended:

Parameter	Value	Reason
`order`	sorted alphabetically	Consistent, reproducible, not driven by the finding
`linewidth`	0.8	Thin box outlines avoid visual heaviness at small multiples
`fliersize`	2	Outlier dots present but small — they are individual data points, not the story

Exclusion: No LOS values excluded. Negative LOS would be an error; `df.describe()` shows min > 0.

Panel 3 — Systolic BP vs Mean Arterial Pressure

Question answered: Which pair of vitals shows the strongest linear relationship (as identified in Panel 4)?
Chart type: Scatter plot with OLS regression line
Week 2 channel rule followed: Position on two common scales is the highest-accuracy channel for encoding two continuous variables simultaneously. Scatter plots directly show the raw bivariate relationship without summary loss.
Parameters chosen and defended:

Parameter	Value	Reason
`alpha`	0.25	10 000 points overplot severely; transparency reveals density without binning
`s`	8	Small markers reduce occlusion at this density
`rasterized`	True	Rasterises the scatter layer to keep the PNG file size reasonable
Subsample	n = 2 000, seed = 42	Rendering 10 000 semi-transparent dots is slow; 2 000 is visually indistinguishable for a correlation this strong. Seed is declared for reproducibility.

Exclusion: The OLS line uses the full cleaned dataset (not the subsample) to avoid bias.

Panel 4 — Correlation Heatmap

Question answered: Which pairs of clinical variables are most strongly correlated?
Chart type: Annotated lower-triangle correlation heatmap
Week 2 channel rule followed: Hue/saturation encodes the signed correlation value in a diverging palette (`RdBu_r`), with red = positive, blue = negative, white = zero. Numerical annotations provide a second encoding so colour is not the sole channel (Week 2, redundant coding). Masking the upper triangle halves the ink while preserving all information (symmetry rule).
Parameters chosen and defended:

Parameter	Value	Reason
`cmap`	`RdBu_r`	Diverging, colour-blind tested, encodes sign; avoids the `Reds`-only mistake of Case 6
`vmin / vmax`	−1 / +1	Maps the full correlation range; prevents weak correlations appearing artificially dark
`annot_kws size`	6.5	Legible at figure scale without crowding a 10×10 cell matrix
`mask`	upper triangle	Removes redundant information; reduces visual clutter

Data Cleaning Summary

#	Problem	Rule	Rows affected	Metric before → after
1	Mixed `admit_date` formats (ISO, UK slash, dd-Mon-YYYY)	`pd.to_datetime(..., format="mixed", dayfirst=True)` — set to NaT if unparseable	Varies (non-zero NaT count)	Not used in any panel
2	Inconsistent ward spelling, not just capitalisation (12 raw labels collapse to 4 true wards)	Normalise to a letters-only lowercase key, map each key to one of 4 canonical names (Title Case alone does not merge “Gen Medicine” with “General Medicine”)	All 10,000 rows	12 raw labels → 4 true wards
3	`heart_rate > 300 bpm`	Threshold: **300 bpm** (task spec: readings above 300 are device faults)	See console output	Heart-rate mean decreases; other vitals unchanged
4	`ward_los_wide.csv` in wide form	`pd.melt()` required for Seaborn (used if loading ward_los_wide.csv)	N/A — used `vitals_10k.csv` directly	—

Symmetry test: The `heart_rate > 300` exclusion removes obvious device artefacts. We apply the same threshold regardless of whether it strengthens or weakens any finding — confirmed by the fact that `heart_rate` does not appear in any panel’s axis, so the exclusion cannot be self-serving.
Appendix note: A version of Panel 1 (systolic histogram) computed before the heart_rate exclusion shows an identical distribution, confirming that the cleaning step did not affect the displayed variable.

Ethics Clause

All data in this assignment (`vitals_10k.csv`) is declared **synthetic** by the course materials. No real patient records were used. Quoted statistics (means, correlations, counts) describe the synthetic dataset only and should not be interpreted as clinical findings. Any exclusion rule applied is disclosed above and would be applied identically if it changed the direction of the finding.

Submitted by: M. Hussain Shah | m.hussain103shah@gmail.com | BSAI-010 Dataset: synthetic, provided by course staff Tools: Python 3.9+, pandas, numpy, matplotlib, seaborn, scipy, vizlib (course toolkit)